Distance-Banded Low-Dimensional Attention: Initial Notes

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1 Motivation

Standard causal self-attention computes pairwise scores

$$s_{ij} = q_i^\top k_j, \qquad j \le i, \tag{1}$$

where all previous tokens interact through the same head dimension. The motivating hypothesis is that nearby tokens often require high-resolution token-level matching, while far-away tokens may only need a lower-dimensional interaction space. If true, attention could spend more representational and computational budget on local context and less on distant context.

The proposed modification is to make the attention score distance-dependent:

$$s_{ij} = (P_{b(i-j)} q_i)^\top (P_{b(i-j)} k_j), \tag{2}$$

where $b(i-j)$ maps the causal distance $i-j$ to a distance band, and P_b projects into a band-specific lower-dimensional subspace. In the pilot experiments, the bands were

$$[0, 128] \mapsto 64, \qquad [129, 512] \mapsto 32, \qquad [513, \infty) \mapsto 16, \tag{3}$$

where the numbers on the right are per-head query/key dimensions. A later aggressive-compression run changed the medium and far dimensions to 16 and 8.

2 Variants Tested

We tested three attention modes:

Baseline. The baseline uses standard causal attention:

$$s_{ij} = q_i^\top k_j. \tag{4}$$

In the current implementation, this uses PyTorch scaled dot-product attention and therefore benefits from FlashAttention.

Distance prefix. This mode uses a single shared query/key projection as in the baseline, but each distance band scores only a prefix of the head vector:

$$s_{ij}^{(b)} = q_{i,1:d_b}^\top k_{j,1:d_b}. \tag{5}$$

This is the simplest version of the idea: far-away interactions use fewer coordinates, but the coordinates are shared with the near-token representation.

Distance per band. This mode gives each band its own learned query/key projections:

$$q_i^{(b)} = W_Q^{(b)} x_i, \qquad k_j^{(b)} = W_K^{(b)} x_j, \qquad s_{ij}^{(b)} = \left(q_i^{(b)}\right)^\top k_j^{(b)}. \tag{6}$$

This allows the model to learn separate matching spaces for near, medium, and far context.

3 Benchmark Setup

The initial language-modeling test used the Parameter Golf-style FineWeb objective with the small GPT configuration currently in the repository:

- sequence length: 1024
- model width: 512
- layers: 9
- attention heads: 8
- KV heads: 4
- training tokens per step: 524,288
- iterations: 800
- validation cadence: every 200 steps
- default distance bands: `128:64,512:32,inf:16`
- aggressive distance bands: `128:64,512:16,inf:8`

The custom distance modes currently use a chunked and checkpointed dense implementation to avoid out-of-memory errors. This makes them runnable for quality experiments, but they do not yet realize the intended speedup. The baseline uses FlashAttention, while the custom modes do not.

4 Initial Results

Mode	Bands	Final train loss	Val loss	Val BPB	Step time (ms)
Baseline	–	2.3065	2.3308	1.3804	864.35
Distance per band	64/32/16	2.3023	2.3290	1.3794	3137.88
Distance prefix	64/32/16	2.3134	2.3413	1.3867	2983.04
Distance per band	64/16/8	2.3112	2.3394	1.3855	3176.41

Table 1: Single-seed 800-step Parameter Golf pilot runs. Lower validation loss and BPB are better.

The main observation is that `distance_per_band` essentially matched the baseline in validation quality in this short run:

$$\Delta\text{BPB} = 1.3794 - 1.3804 = -0.0010. \tag{7}$$

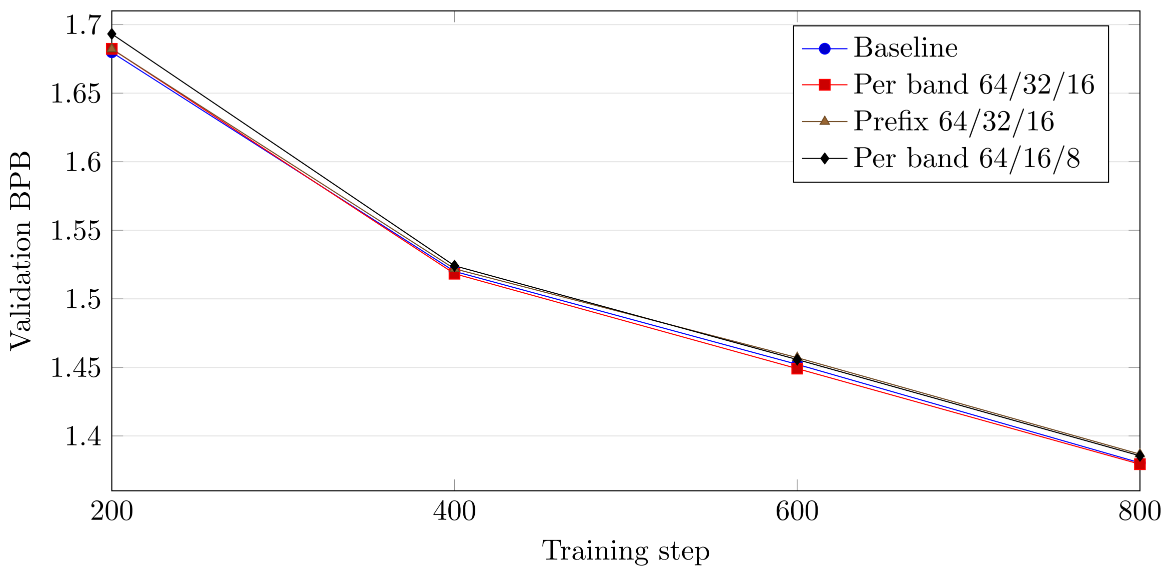

Figure 1: Validation BPB trajectories after the initial step-0 evaluation. The curves are close; the final differences are easier to interpret as step-equivalent lags.

This difference is small enough that it should be treated as noise until tested across more seeds, but it is encouraging: the lower-dimensional far-context interaction did not obviously hurt the model.

The `distance_prefix` variant was worse:

$$\Delta\text{BPB} = 1.3867 - 1.3804 = +0.0063. \tag{8}$$

This suggests that simply truncating a shared query/key vector may be too restrictive. The model appears to benefit from learning separate distance-specific matching spaces.

The more aggressive `distance_per_band` run used `128:64,512:16,inf:8`. It landed between the mild per-band run and the prefix run:

$$1.3855 - 1.3804 = +0.0051 \tag{9}$$

BPB worse than baseline, and

$$1.3855 - 1.3794 = +0.0061 \tag{10}$$

BPB worse than the `64/32/16` per-band setting. This suggests that the far-context dimensions can likely be reduced, but the `16/8` medium/far setting may be too aggressive at this model scale and training horizon.

4.1 Step-Equivalent Interpretation

The raw BPB differences are small, so a useful scale is to compare them against the slope of the baseline learning curve. From step 600 to step 800, the baseline improves from 1.4522 BPB to 1.3804 BPB:

$$\frac{1.4522 - 1.3804}{200} \approx 0.000359 \quad \text{BPB/step.} \tag{11}$$

Using this final-segment slope, each method's final BPB gap can be converted into an approximate number of baseline training steps.

Mode	Final BPB gap vs baseline	Step-equivalent lag
Distance per band 64/32/16	-0.0010	≈ 3 steps ahead
Distance per band 64/16/8	$+0.0051$	≈ 14 steps behind
Distance prefix 64/32/16	$+0.0063$	≈ 18 steps behind

Table 2: Approximate final gaps measured in units of late-stage baseline training progress. Negative BPB gap means better than baseline.

This framing suggests that the mild per-band model is effectively tied with baseline at this resolution, while the aggressive per-band and prefix variants lag by only a small fraction of the 800-step run. This is not a substitute for multi-seed statistics, but it gives the final differences a more intuitive scale.

5 Interpretation

The initial result supports a narrow but useful claim: distance-dependent query/key dimensionality can preserve short-run language-model quality, at least when implemented with learned per-band projections. It does not yet support a speed claim, because the current custom implementation is much slower than the FlashAttention baseline:

$$\frac{3137.88}{864.35} \approx 3.63. \tag{12}$$

Thus the current conclusion is:

> The architectural idea is not obviously quality-destructive, and the per-band projection structure looks more promising than prefix truncation. However, a real efficiency result requires a fused or genuinely banded attention implementation that avoids materializing dense score matrices.

6 Next Experiments

Natural next tests are:

1. Run more seeds at the same 800-step setting to estimate noise.
2. Sweep intermediate compression settings, e.g. `128:64,512:24,inf:12`.
3. Compare parameter-matched versions, since `distance_per_band` changes the query/key parameterization.
4. Implement a fused/windowed kernel or block-sparse attention path so the method can be tested for actual speed and memory improvements.